

EDUCATION

- **University POLITEHNICA of Bucharest** Bucharest, Romania
BSc in Systems Engineering *Sept. 2022 – July 2026*
- **”Alexandru Vlahuță” National College** Râmnicu Sărat, Romania
Bachelor’s Degree, Mathematics and Computer Science *Sept. 2018 – July 2022*

EXPERIENCE

- **Mathnasium** Bucharest, Romania
Mathematics Instructor *Oct. 2025 - June 2026*
 - **Personalized Instruction:** Delivered tailored math instruction to students, utilizing the Mathnasium Method to strengthen numerical fluency and problem-solving skills.
 - **Student Assessment:** Evaluated individual student progress through regular diagnostic testing and adjusted learning plans to meet specific academic milestones.
 - **Curriculum Management:** Managed educational materials and ensured a productive learning environment, fostering student engagement and confidence in mathematics.
 - **Stakeholder Communication:** Communicated complex mathematical progress and behavioral insight to parents and guardians to ensure alignment on student goals.

PROJECTS

- **Personal Website:** Designed and developed a responsive, professional portfolio website to showcase engineering projects and technical skills.
- **Adaptive Math Tutoring Platform - Diploma Project — UPB 2026 (GitHub):** Developed a web-based adaptive math tutoring platform for Romanian gymnasium students (grades 4–8), integrating multimodal answer evaluation (text, photo, digital whiteboard) with an adaptive difficulty algorithm formulated as a closed-loop control system. Implemented Retrieval-Augmented Generation (RAG) over digitized Romanian textbooks using Chroma vector database and LangChain to ground AI feedback in the official MEC curriculum. Built a dedicated teacher module with automated gap detection, flagging students with 3+ consecutive errors per topic. Engineered the backend in Flask with SQLAlchemy and PostgreSQL, integrated Google Gemini 2.5 Flash for multimodal reasoning.
- **Emergency Call and Intervention Management System (GitHub):** Developed a centralized platform for real-time management of emergency calls and rescue operators. Designed and implemented a relational database with interlinked tables (Dispatchers, Callers, Calls, etc.) to ensure data integrity and complex associations. Created a web-based interface for dispatchers to perform full CRUD (Create, Read, Update, Delete) operations and manage intervention workflows efficiently. Engineered complex SQL queries to generate statistical reports, such as identifying high-priority areas and monitoring dispatcher performance using advanced JOINS and subqueries.
- **Fashion MNIST Image Classification (GitHub):** Developed a machine learning model to classify fashion items from Fashion MNIST dataset. Implemented a comprehensive data preprocessing pipeline, including feature scaling and dimensionality reduction to 100 components to optimize computational efficiency. Trained and optimized a Random Forest Classifier for hyperparameter tuning to improve model robustness and prevent overfitting. Conducted performance evaluation using a validation subset and automated the prediction generation process for test datasets.

ACHIEVEMENTS AND RESPONSIBILITIES

- **Educational Leadership:** Managed personalized learning plans for students as a Math Instructor at Mathnasium, ensuring academic progress and numerical fluency through tailored instruction.
- **Performance Accountability:** Monitored and evaluated individual student progress at Mathnasium.
- **Professional Competence Certificate in Computer Science:** Certified in designing software structures, databases, and web pages, as well as customizing applications for client-specific information systems.
- **Oracle Academy Award for Database Programming With SQL:** Completed the Final Exam on 13th May 2022.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, Matlab, C, Java, SQL, PHP.
- **Web Development:** Flask, HTML, CSS, JavaScript, KaTeX, Chart.js.
- **Databases:** PostgreSQL, SQLite, Chroma.
- **AI & Machine Learning:** Google Gemini, LangChain, Retrieval-Augmented Generation.
- **Tools & Platforms:** Git, Linux, LaTeX, Simulink.

ADDITIONAL SKILLS

- **Languages:** Romanian (native), English (advanced).
- **Soft Skills:** Mathematics mentoring for gymnasaium students, effective communication, analytical thinking.